

MySQL Solutions for All Exercises

```
mysql> CREATE DATABASE eventportal;  
Query OK, 1 row affected (0.01 sec)
```

```
mysql> use eventportal  
Database changed  
mysql> |
```

```
mysql> CREATE TABLE Users(  
->     user_id INT AUTO_INCREMENT PRIMARY KEY,  
->     full_name VARCHAR(100) NOT NULL,  
->     email VARCHAR(100) UNIQUE NOT NULL,  
->     city VARCHAR(100) NOT NULL,  
->     registration_date DATE NOT NULL  
-> );  
Query OK, 0 rows affected (0.03 sec)
```

```
mysql>  
mysql> CREATE TABLE Events(  
->     event_id INT AUTO_INCREMENT PRIMARY KEY,  
->     title VARCHAR(200) NOT NULL,  
->     description TEXT,  
->     city VARCHAR(100) NOT NULL,  
->     start_date DATETIME NOT NULL,  
->     end_date DATETIME NOT NULL,  
->     status ENUM('upcoming','completed','cancelled'),  
->     organizer_id INT,  
->     FOREIGN KEY (organizer_id)  
->     REFERENCES Users(user_id)  
-> );  
Query OK, 0 rows affected (0.02 sec)
```

```
mysql>  
mysql> CREATE TABLE Sessions(  
->     session_id INT AUTO_INCREMENT PRIMARY KEY,  
->     event_id INT,  
->     title VARCHAR(200) NOT NULL,  
->     speaker_name VARCHAR(100) NOT NULL,  
->     start_time DATETIME NOT NULL,  
->     end_time DATETIME NOT NULL,  
->     FOREIGN KEY (event_id)  
->     REFERENCES Events(event_id)  
-> );  
Query OK, 0 rows affected (0.02 sec)
```

```
mysql>
mysql> CREATE TABLE Registrations(
->     registration_id INT AUTO_INCREMENT PRIMARY KEY,
->     user_id INT,
->     event_id INT,
->     registration_date DATE NOT NULL,
->     FOREIGN KEY (user_id)
->     REFERENCES Users(user_id),
->     FOREIGN KEY (event_id)
->     REFERENCES Events(event_id)
-> );
Query OK, 0 rows affected (0.03 sec)
```

```
mysql>
mysql> CREATE TABLE Feedback(
->     feedback_id INT AUTO_INCREMENT PRIMARY KEY,
->     user_id INT,
->     event_id INT,
->     rating INT CHECK(rating BETWEEN 1 AND 5),
->     comments TEXT,
->     feedback_date DATE NOT NULL,
->     FOREIGN KEY(user_id)
->     REFERENCES Users(user_id),
->     FOREIGN KEY(event_id)
->     REFERENCES Events(event_id)
-> );
Query OK, 0 rows affected (0.03 sec)
```

```
mysql>
mysql> CREATE TABLE Resources(
->     resource_id INT AUTO_INCREMENT PRIMARY KEY,
->     event_id INT,
->     resource_type ENUM('pdf','image','link'),
->     resource_url VARCHAR(255) NOT NULL,
->     uploaded_at DATETIME NOT NULL,
->     FOREIGN KEY(event_id)
->     REFERENCES Events(event_id)
-> );
Query OK, 0 rows affected (0.03 sec)
```

Inserting values into the tables

```
mysql> INSERT INTO Users VALUES
-> (1,'Alice Johnson','alice@example.com','New York','2024-12-01'),
-> (2,'Bob Smith','bob@example.com','Los Angeles','2024-12-05'),
-> (3,'Charlie Lee','charlie@example.com','Chicago','2024-12-10'),
-> (4,'Diana King','diana@example.com','New York','2025-01-15'),
-> (5,'Ethan Hunt','ethan@example.com','Los Angeles','2025-02-01');
Query OK, 5 rows affected (0.03 sec)
Records: 5  Duplicates: 0  Warnings: 0
```

```
mysql> INSERT INTO Events VALUES
-> (1,'Tech Innovators Meetup',
-> 'A meetup for tech enthusiasts.',
-> 'New York',
-> '2025-06-10 10:00:00',
-> '2025-06-10 16:00:00',
-> 'upcoming',
-> 1),
->
-> (2,'AI & ML Conference',
-> 'Conference on AI and ML advancements.',
-> 'Chicago',
-> '2025-05-15 09:00:00',
-> '2025-05-15 17:00:00',
-> 'completed',
-> 3),
->
-> (3,'Frontend Development Bootcamp',
-> 'Hands-on training on frontend tech.',
-> 'Los Angeles',
-> '2025-07-01 10:00:00',
-> '2025-07-03 16:00:00',
-> 'upcoming',
-> 2);
Query OK, 3 rows affected (0.01 sec)
Records: 3 Duplicates: 0 Warnings: 0
```

```
mysql> INSERT INTO Sessions VALUES
-> (1,1,'Opening Keynote','Dr. Tech',
-> '2025-06-10 10:00:00',
-> '2025-06-10 11:00:00'),
->
-> (2,1,'Future of Web Dev','Alice Johnson',
-> '2025-06-10 11:15:00',
-> '2025-06-10 12:30:00'),
->
-> (3,2,'AI in Healthcare','Charlie Lee',
-> '2025-05-15 09:30:00',
-> '2025-05-15 11:00:00'),
->
-> (4,3,'Intro to HTML5','Bob Smith',
-> '2025-07-01 10:00:00',
-> '2025-07-01 12:00:00');
Query OK, 4 rows affected (0.01 sec)
Records: 4 Duplicates: 0 Warnings: 0
```

```
mysql> INSERT INTO Registrations VALUES
-> (1,1,1,'2025-05-01'),
-> (2,2,1,'2025-05-02'),
-> (3,3,2,'2025-04-30'),
-> (4,4,2,'2025-04-28'),
-> (5,5,3,'2025-06-15');
Query OK, 5 rows affected (0.01 sec)
Records: 5 Duplicates: 0 Warnings: 0
```

```
mysql> INSERT INTO Feedback VALUES
-> (1,3,2,4,'Great insights!','2025-05-16'),
-> (2,4,2,5,'Very informative.','2025-05-16'),
-> (3,2,1,3,'Could be better.','2025-06-11');
Query OK, 3 rows affected (0.01 sec)
Records: 3 Duplicates: 0 Warnings: 0
```

```
mysql> INSERT INTO Resources VALUES
-> (1,1,'pdf',
-> 'https://portal.com/resources/tech_meetup_agenda.pdf',
-> '2025-05-01 10:00:00'),
->
-> (2,2,'image',
-> 'https://portal.com/resources/ai_poster.jpg',
-> '2025-04-20 09:00:00'),
->
-> (3,3,'link',
-> 'https://portal.com/resources/html5_docs',
-> '2025-06-25 15:00:00');
Query OK, 3 rows affected (0.01 sec)
Records: 3 Duplicates: 0 Warnings: 0
```

```
mysql> select * from events;
+-----+-----+-----+-----+-----+-----+-----+-----+
| event_id | title                | description                | city      | start_date | end_date | status | organizer_id |
+-----+-----+-----+-----+-----+-----+-----+-----+
| 1        | Tech Innovators Meetup | A meetup for tech enthusiasts. | New York | 2025-06-10 10:00:00 | 2025-06-10 16:00:00 | upcoming | 1            |
| 2        | AI & ML Conference    | Conference on AI and ML advancements. | Chicago  | 2025-05-15 09:00:00 | 2025-05-15 17:00:00 | completed | 3            |
| 3        | Frontend Development Bootcamp | Hands-on training on frontend tech. | Los Angeles | 2025-07-01 10:00:00 | 2025-07-03 16:00:00 | upcoming | 2            |
+-----+-----+-----+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)

mysql> select * from feedback;
+-----+-----+-----+-----+-----+-----+
| feedback_id | user_id | event_id | rating | comments          | feedback_date |
+-----+-----+-----+-----+-----+-----+
| 1           | 3       | 2        | 4       | Great insights!   | 2025-05-16    |
| 2           | 4       | 2        | 5       | Very informative. | 2025-05-16    |
| 3           | 2       | 1        | 3       | Could be better.  | 2025-06-11    |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)

mysql> select * from registrations;
+-----+-----+-----+-----+
| registration_id | user_id | event_id | registration_date |
+-----+-----+-----+-----+
| 1              | 1       | 1        | 2025-05-01        |
| 2              | 2       | 1        | 2025-05-02        |
| 3              | 3       | 2        | 2025-04-30        |
| 4              | 4       | 2        | 2025-04-28        |
| 5              | 5       | 3        | 2025-06-15        |
+-----+-----+-----+-----+
5 rows in set (0.00 sec)

mysql> select * from resources;
+-----+-----+-----+-----+-----+
| resource_id | event_id | resource_type | resource_url | uploaded_at |
+-----+-----+-----+-----+-----+
| 1           | 1        | pdf          | https://portal.com/resources/tech_meetup_agenda.pdf | 2025-05-01 10:00:00 |
| 2           | 2        | image       | https://portal.com/resources/ai_poster.jpg          | 2025-04-20 09:00:00 |
| 3           | 3        | link        | https://portal.com/resources/html5_docs              | 2025-06-25 15:00:00 |
+-----+-----+-----+-----+-----+
3 rows in set (0.01 sec)

mysql> select * from sessions;
+-----+-----+-----+-----+-----+-----+
| session_id | event_id | title                | speaker_name | start_time | end_time |
+-----+-----+-----+-----+-----+-----+
| 1          | 1        | Opening Keynote      | Dr. Tech    | 2025-06-10 10:00:00 | 2025-06-10 11:00:00 |
| 2          | 2        | Future of Web Dev    | Alice Johnson | 2025-06-10 11:15:00 | 2025-06-10 12:30:00 |
| 3          | 2        | AI in Healthcare     | Charlie Lee  | 2025-05-15 09:30:00 | 2025-05-15 11:00:00 |
| 4          | 3        | Intro to HTML5       | Bob Smith   | 2025-07-01 10:00:00 | 2025-07-01 12:00:00 |
+-----+-----+-----+-----+-----+-----+
4 rows in set (0.00 sec)

mysql> select * from users;
+-----+-----+-----+-----+-----+
| user_id | full_name | email                | city      | registration_date |
+-----+-----+-----+-----+-----+
| 1       | Alice Johnson | alice@example.com    | New York | 2024-12-01        |
| 2       | Bob Smith    | bob@example.com      | Los Angeles | 2024-12-05        |
| 3       | Charlie Lee  | charlie@example.com  | Chicago   | 2024-12-10        |
| 4       | Diana King   | diana@example.com    | New York  | 2025-01-15        |
| 5       | Ethan Hunt   | ethan@example.com    | Los Angeles | 2025-02-01        |
+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)
```

EXERCISES:

1. User Upcoming Events:

```
mysql> SELECT u.full_name, e.title, e.city, e.start_date
-> FROM Users u
-> JOIN Registrations r ON u.user_id = r.user_id
-> JOIN Events e ON r.event_id = e.event_id
-> WHERE e.status = 'upcoming'
-> AND u.city = e.city
-> ORDER BY e.start_date;
```

full_name	title	city	start_date
Alice Johnson	Tech Innovators Meetup	New York	2025-06-10 10:00:00
Ethan Hunt	Frontend Development Bootcamp	Los Angeles	2025-07-01 10:00:00

2 rows in set (0.00 sec)

```
mysql> |
```

2. Top Rated Events:

```
mysql> SELECT e.title,
->         AVG(f.rating) AS avg_rating,
->         COUNT(f.feedback_id) AS total_feedback
-> FROM Events e
-> JOIN Feedback f ON e.event_id = f.event_id
-> GROUP BY e.event_id, e.title
-> HAVING COUNT(f.feedback_id) >= 10
-> ORDER BY avg_rating DESC;
```

Empty set (0.01 sec)

3. Inactive Users:

```
mysql> SELECT *
-> FROM Users
-> WHERE user_id NOT IN
-> (
->     SELECT DISTINCT user_id
->     FROM Registrations
->     WHERE registration_date >= CURDATE() - INTERVAL 90 DAY
-> );
```

user_id	full_name	email	city	registration_date
1	Alice Johnson	alice@example.com	New York	2024-12-01
2	Bob Smith	bob@example.com	Los Angeles	2024-12-05
3	Charlie Lee	charlie@example.com	Chicago	2024-12-10
4	Diana King	diana@example.com	New York	2025-01-15
5	Ethan Hunt	ethan@example.com	Los Angeles	2025-02-01

5 rows in set (0.01 sec)

4. Peak session Hours

```
mysql> SELECT e.title,
->         COUNT(s.session_id) AS sessions_between_10_12
-> FROM Events e
-> LEFT JOIN Sessions s
-> ON e.event_id = s.event_id
-> WHERE HOUR(s.start_time) BETWEEN 10 AND 12
-> GROUP BY e.title;
```

title	sessions_between_10_12
Tech Innovators Meetup	2
Frontend Development Bootcamp	1

2 rows in set (0.00 sec)

5. Most Active cities

```
mysql> SELECT u.city,  
-> COUNT(DISTINCT r.registration_id) AS registrations  
-> FROM Users u  
-> JOIN Registrations r  
-> ON u.user_id = r.user_id  
-> GROUP BY u.city  
-> ORDER BY registrations DESC  
-> LIMIT 5;
```

city	registrations
Los Angeles	2
New York	2
Chicago	1

3 rows in set (0.00 sec)

6. Event Resource Summary

```
mysql> SELECT e.title,  
-> COUNT(r.resource_id) AS total_resources  
-> FROM Events e  
-> LEFT JOIN Resources r  
-> ON e.event_id = r.event_id  
-> GROUP BY e.title;
```

title	total_resources
Tech Innovators Meetup	1
AI & ML Conference	1
Frontend Development Bootcamp	1

3 rows in set (0.00 sec)

7. Low Feedback Alerts

```
mysql> SELECT u.full_name,  
-> e.title,  
-> f.rating,  
-> f.comments  
-> FROM Feedback f  
-> JOIN Users u ON f.user_id = u.user_id  
-> JOIN Events e ON f.event_id = e.event_id  
-> WHERE f.rating < 3;
```

full_name	title	rating	comments
Alice Johnson	Tech Innovators Meetup	2	Poor organization

1 row in set (0.00 sec)

8. Sessions per Upcoming Events

```
mysql> SELECT e.title,  
-> COUNT(s.session_id) AS session_count  
-> FROM Events e  
-> LEFT JOIN Sessions s  
-> ON e.event_id = s.event_id  
-> WHERE e.status='upcoming'  
-> GROUP BY e.title;
```

title	session_count
Tech Innovators Meetup	2
Frontend Development Bootcamp	1

2 rows in set (0.00 sec)

9. Organizer Event Summary

```
mysql> SELECT u.full_name,  
->      e.status,  
->      COUNT(*) AS total_events  
-> FROM Users u  
-> JOIN Events e  
-> ON u.user_id=e.organizer_id  
-> GROUP BY u.full_name,e.status;
```

full_name	status	total_events
Alice Johnson	upcoming	1
Charlie Lee	completed	1
Bob Smith	upcoming	1

3 rows in set (0.00 sec)

10. Feedback Gap

```
mysql> SELECT e.title  
-> FROM Events e  
-> JOIN Registrations r  
-> ON e.event_id=r.event_id  
-> LEFT JOIN Feedback f  
-> ON e.event_id=f.event_id  
-> WHERE f.feedback_id IS NULL  
-> GROUP BY e.title;
```

title
Frontend Development Bootcamp

1 row in set (0.00 sec)

11. Daily New User Count

```
mysql> SELECT registration_date,  
->      COUNT(*) AS new_users  
-> FROM Users  
-> GROUP BY registration_date  
-> ORDER BY registration_date;
```

registration_date	new_users
2024-12-01	1
2024-12-05	1
2024-12-10	1
2025-01-15	1
2025-02-01	1

5 rows in set (0.00 sec)

12. Event with Maximum Sessions

```
mysql> SELECT e.title,
-> COUNT(s.session_id) AS total_sessions
-> FROM Events e
-> JOIN Sessions s
-> ON e.event_id=s.event_id
-> GROUP BY e.event_id,e.title
-> HAVING COUNT(s.session_id)=
-> (
-> SELECT MAX(session_count)
-> FROM
-> (
-> SELECT COUNT(*) session_count
-> FROM Sessions
-> GROUP BY event_id
-> ) x
-> );
```

title	total_sessions
Tech Innovators Meetup	2

1 row in set (0.01 sec)

13. Average rating Per city

```
mysql> SELECT e.city,
-> AVG(f.rating) AS avg_rating
-> FROM Events e
-> JOIN Feedback f
-> ON e.event_id=f.event_id
-> GROUP BY e.city;
```

city	avg_rating
New York	2.5000
Chicago	4.5000

2 rows in set (0.00 sec)

14. Most Registered Events

```
mysql> SELECT e.title,
-> COUNT(r.registration_id) AS registrations
-> FROM Events e
-> JOIN Registrations r
-> ON e.event_id=r.event_id
-> GROUP BY e.title
-> ORDER BY registrations DESC
-> LIMIT 3;
```

title	registrations
Tech Innovators Meetup	2
AI & ML Conference	2
Frontend Development Bootcamp	1

3 rows in set (0.00 sec)

15. Event Session Time Conflict

```
mysql> SELECT
-> s1.event_id,
-> s1.title AS session1,
-> s2.title AS session2
-> FROM Sessions s1
-> JOIN Sessions s2
-> ON s1.event_id=s2.event_id
-> AND s1.session_id < s2.session_id
-> AND s1.start_time < s2.end_time
-> AND s1.end_time > s2.start_time;
```

event_id	session1	session2
1	Opening Keynote	React Workshop
1	Future of Web Dev	React Workshop

2 rows in set (0.00 sec)

16. Unregistered Active Users

```
mysql> SELECT *
-> FROM Users u
-> WHERE u.registration_date >= CURDATE() - INTERVAL 30 DAY
-> AND u.user_id NOT IN
-> (
-> SELECT user_id
-> FROM Registrations
-> );
Empty set (0.00 sec)
```

17. Multi Session Speakers

```
mysql> SELECT speaker_name,
-> COUNT(*) AS sessions_handled
-> FROM Sessions
-> GROUP BY speaker_name
-> HAVING COUNT(*) > 1;
Empty set (0.00 sec)
```

18. Resource Availability Check

```
mysql> ^C
mysql> SELECT e.title
-> FROM Events e
-> LEFT JOIN Resources r
-> ON e.event_id=r.event_id
-> WHERE r.resource_id IS NULL;
Empty set (0.00 sec)

mysql> INSERT INTO Events
-> (title,description,city,start_date,end_date,status,organizer_id)
-> VALUES
-> (
-> 'Cloud Computing Workshop',
-> 'Workshop on cloud technologies',
-> 'Chicago',
-> '2025-08-10 10:00:00',
-> '2025-08-10 16:00:00',
-> 'upcoming',
-> 1
-> );
Query OK, 1 row affected (0.01 sec)

mysql> SELECT e.title
-> FROM Events e
-> LEFT JOIN Resources r
-> ON e.event_id=r.event_id
-> WHERE r.resource_id IS NULL;

+-----+
| title |
+-----+
| Cloud Computing Workshop |
+-----+
1 row in set (0.00 sec)

mysql> |
```

19. Completed Events with Feedback Summary

```
mysql> SELECT e.title,
-> COUNT(DISTINCT r.registration_id) AS registrations,
-> AVG(f.rating) AS avg_rating
-> FROM Events e
-> LEFT JOIN Registrations r
-> ON e.event_id=r.event_id
-> LEFT JOIN Feedback f
-> ON e.event_id=f.event_id
-> WHERE e.status='completed'
-> GROUP BY e.title;

+-----+-----+-----+
| title | registrations | avg_rating |
+-----+-----+-----+
| AI & ML Conference | 2 | 4.5000 |
+-----+-----+-----+
1 row in set (0.00 sec)
```

20. User Engagement Index

```
mysql> SELECT u.full_name,
-> COUNT(DISTINCT r.event_id) AS events_attended,
-> COUNT(DISTINCT f.feedback_id) AS feedbacks_given
-> FROM Users u
-> LEFT JOIN Registrations r
-> ON u.user_id=r.user_id
-> LEFT JOIN Feedback f
-> ON u.user_id=f.user_id
-> GROUP BY u.user_id,u.full_name;

+-----+-----+-----+
| full_name | events_attended | feedbacks_given |
+-----+-----+-----+
| Alice Johnson | 1 | 1 |
| Bob Smith | 1 | 1 |
| Charlie Lee | 1 | 1 |
| Diana King | 1 | 1 |
| Ethan Hunt | 1 | 0 |
+-----+-----+-----+
5 rows in set (0.00 sec)
```

21. Top feedback providers

```
mysql> SELECT u.full_name,
->      COUNT(f.feedback_id) AS feedback_count
-> FROM Users u
-> JOIN Feedback f
-> ON u.user_id=f.user_id
-> GROUP BY u.full_name
-> ORDER BY feedback_count DESC
-> LIMIT 5;
```

full_name	feedback_count
Alice Johnson	1
Bob Smith	1
Charlie Lee	1
Diana King	1

4 rows in set (0.00 sec)

22. Duplicate Registrations Check

```
mysql> SELECT user_id,
->      event_id,
->      COUNT(*) AS duplicate_count
-> FROM Registrations
-> GROUP BY user_id,event_id
-> HAVING COUNT(*) > 1;
Empty set (0.00 sec)
```

23. Registration trend

```
mysql> SELECT
->      YEAR(registration_date) AS year,
->      MONTH(registration_date) AS month,
->      COUNT(*) AS registrations
-> FROM Registrations
-> WHERE registration_date >= CURDATE() - INTERVAL 12 MONTH
-> GROUP BY YEAR(registration_date),
->      MONTH(registration_date)
-> ORDER BY year,month;
```

year	month	registrations
2025	6	1

1 row in set (0.00 sec)

24. Average Session Duration Per Event

```
mysql> SELECT e.title,
->      AVG(
->      TIMESTAMPDIFF
->      (
->      MINUTE,
->      s.start_time,
->      s.end_time,
->      )
->      ) AS avg_duration_minutes
-> FROM Events e
-> JOIN Sessions s
-> ON e.event_id=s.event_id
-> GROUP BY e.title;
```

title	avg_duration_minutes
Tech Innovators Meetup	65.0000
AI & ML Conference	90.0000
Frontend Development Bootcamp	120.0000

3 rows in set (0.00 sec)

25. Events without Sessions

```
mysql> SELECT e.title
-> FROM Events e
-> LEFT JOIN Sessions s
-> ON e.event_id=s.event_id
-> WHERE s.session_id IS NULL;
```

title
Cloud Computing Workshop

1 row in set (0.00 sec)